SOLUTION REQUIREMENTS

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine (EALSE)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the functional and non-functional requirements for the Emotion-Aware Learning Support Engine (EALSE). Functional requirements describe what the system should do, while non-functional requirements specify quality, performance, security and usability expectations.

Step 1 : Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Role selection (Learner / Educator) sets session context; no separate login required for the prototype.	Medium
2	Authorization Levels	Role-based response tailoring for Learner vs. Educator/TA; educators additionally access the analytics dashboard.	High
3	External Interfaces	Integrates with the Google Gemini API (gemini-2.5-flash) for guidance generation and HuggingFace Transformers for DistilBERT inference.	High
4	Transaction Processing	Accept free-text input, run dual-model emotion inference (BiLSTM + DistilBERT), compute the final prediction, and generate AI guidance.	High
5	Reporting	Log every interaction to data/interactions.csv and display an analytics dashboard with emotion distribution and confidence trends.	High
6	Business Rules	Input limited to 2,000 characters; a mixed emotion is flagged only when the secondary class confidence is 15% or higher.	Medium
7	Compliance	Store GEMINI_API_KEY only in .env (never committed to version control); no PII beyond typed text is persisted.	High

S.No	Requirement Category	Requirement Description	Priority
8	Other	Provide guidance regeneration, side-by-side model comparison cards, and graceful setup instructions when models or the API key are missing.	Medium

Step 2 : Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast dual-model inference and guidance generation.	End-to-end response within 5 seconds on Intel i3 / 4GB RAM (excluding Gemini network latency).
2	Scalability	Support multiple concurrent Streamlit sessions.	Handles typical classroom-size concurrent usage without noticeable degradation.
3	Security & Data Privacy	Protect API keys and logged interaction data.	API key stored only in .env; local-only CSV logs with no cloud persistence.
4	Reliability & Availability	System should remain available and degrade gracefully.	Automatic Gemini model fallback on quota errors; app does not crash when model files are missing.
5	Usability & Accessibility	Responsive, clear interface with emotion visualizations.	Usable on common desktop browsers; result cards and charts render clearly at standard screen widths.
6	Other	Maintainability and portability of code.	Modular Python codebase (preprocessing, model, predict, train, app) with version-independent NPZ weight storage.